

Zepto Retail Data Analysis

1. Project Overview

This project analyzes retail product data from Zepto using PostgreSQL. The objective is to perform data cleaning, exploratory data analysis, and business analytics to derive meaningful insights related to pricing, discounts, inventory, and product categories.

2. Dataset Summary

The dataset contains product-level information including:

- SKU ID
- Product Name
- Category
- Maximum Retail Price (MRP)
- Discount Percentage
- Discounted Selling Price
- Available Quantity
- Product Weight
- Stock Availability
- Quantity

The data was imported into PostgreSQL for analysis.

3. Tools Used

- PostgreSQL
 - pgAdmin 4
 - SQL
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4. Data Exploration

Total Number of Products

Statement:

Calculated the total number of products available in the dataset.

Output:

	count 
1	3731

Sample Data Inspection

Statement:

Displayed the first few records to understand the dataset structure.

Output:

Showing rows: 1 to 10  Page No: 1 of 1    										
	skuid 	category 	name 	mrp 	discountpercent 	availablequantity 	discountedsellingprice 	weightingms 	outofstock 	quantity 
1	1	Fruits & Vegetables	Onion	25.00	16.00	3	21.00	1000	false	1
2	2	Fruits & Vegetables	Tomato Hybrid	42.00	16.00	3	35.00	1000	false	1
3	768	Munchies	Everest Sabji Masala	30.00	3.00	6	29.00	50	false	50
4	842	Munchies	Popular Essentials Ajwain W...	50.00	12.00	2	44.00	100	false	100
5	1272	Beverages	ID Curd Cup	60.00	0.00	6	60.00	400	false	400
6	1543	Packaged Food	Kellogg's Corn Flakes	425.00	9.00	1	383.00	1000	false	1
7	2132	Ice Cream & Desserts	Top Ramen Masala	91.00	9.00	0	82.00	560	true	560
8	2544	Meats, Fish & Eggs	Arden Eggs White	97.00	0.00	4	97.00	696	false	12
9	2655	Biscuits	Parle Rusk Elaichi	48.00	2.00	6	47.00	300	false	300
10	2788	Personal Care	Gillette Sensitive Shaving Fo...	225.00	6.00	4	210.00	418	false	418

Missing Value Check

Statement:

Verified whether any important columns contain missing values.

Output:

skuid [PK] integer	category character varying (120)	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)	availablequantity integer	discountedsellingprice numeric (8,2)	weightingms integer	outofstock boolean	quantity integer
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Product Categories

Statement:

Retrieved all unique product categories available in the dataset.

Output:

	category character varying (120)
1	Beverages
2	Biscuits
3	Chocolates & Candies
4	Cooking Essentials
5	Dairy, Bread & Batter
6	Fruits & Vegetables
7	Health & Hygiene
8	Home & Cleaning
9	Ice Cream & Desserts
10	Meats, Fish & Eggs
11	Munchies
12	Paan Corner
13	Packaged Food
14	Personal Care

Stock Availability

Statement:

Calculated the number of products that are currently in stock and out of stock.

Output:

	outofstock boolean 🔒	count bigint 🔒
1	false	3278
2	true	453

5. Data Cleaning

Invalid Price Removal

Statement:

Removed products with invalid MRP values to improve data quality.

Price Standardization

Statement:

Converted price values into Indian Rupees for easier interpretation and analysis.

Output:

	mrp numeric (8,2) 🔒	discountedsellingprice numeric (8,2) 🔒
1	0.32	0.27
2	0.54	0.45
3	1.77	1.49
4	0.32	0.27
5	0.30	0.25
6	1.89	1.59
7	0.30	0.25
8	0.23	0.19
9	0.35	0.29
10	0.35	0.29
11	0.35	0.29
12	0.42	0.35
13	0.20	0.17
14	0.29	0.24
15	0.35	0.29
16	0.38	0.32
17	0.89	0.75
18	0.54	0.45
19	0.70	0.59
20	0.42	0.35

6. Business Analysis

Top 10 Highest Discounted Products

Statement:

Identified products offering the highest discount percentages.

Output:

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Chocolate Wafers	0.45	51.00
2	Dukes Waffy Strawberry Wafers	0.45	51.00
3	Dukes Waffy Orange Wafers	0.45	51.00
4	Dukes Waffy Strawberry Roll	1.50	50.00
5	RRO Fresh Ricotta	2.75	50.00
6	RRO Mozzarella Pizza Cheese	2.75	50.00
7	Moi Soi Manchurian Sauce - For Stir Fry Marinade Spread & ...	2.80	50.00
8	RRO Sour Cream	2.75	50.00
9	RRO Mascarpone Cheese	3.55	50.00
10	Chef's Basket Durum Wheat Elbow Pasta	1.60	50.00

Insight:

These products provide the maximum savings and can be highlighted during promotional campaigns.

Premium Products Currently Out of Stock

Statement:

Retrieved expensive products that are currently unavailable.

Insight:

These products may require inventory replenishment to avoid revenue loss.

Estimated Revenue by Category

Statement:

Calculated the estimated revenue generated by each product category.

Output:

	category character varying (120) 🔒	total_revenue numeric 🔒
1	Fruits & Vegetables	108.46
2	Meats, Fish & Eggs	206.93
3	Biscuits	250.19
4	Beverages	550.51
5	Dairy, Bread & Batter	550.51
6	Health & Hygiene	641.80
7	Home & Cleaning	1226.61
8	Ice Cream & Desserts	2243.85
9	Chocolates & Candies	2243.85
10	Packaged Food	2243.85
11	Personal Care	2708.49
12	Paan Corner	2708.49
13	Munchies	3373.69
14	Cooking Essentials	3373.69

Insight:

The analysis identifies categories contributing the highest share of potential revenue.

High-Value Products with Low Discounts

Statement:

Identified expensive products that receive minimal discounts.

Insight:

These products maintain premium pricing while offering limited promotional benefits.

Categories with Highest Average Discounts

Statement:

Compared average discount percentages across product categories.

Output:

	category character varying (120)	avg_discount numeric
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

Insight:

This helps identify categories where aggressive discounting strategies are applied.

Best Value Products (Price per Gram)

Statement:

Calculated price per gram to determine products offering the best value for money.

Output:

	name character varying (150)	weightingms integer	discountedsellingprice numeric (8,2)	price_per_gram numeric
1	Nupur Heena Mehendi	500	1.99	0.00
2	24 Mantra Mustard Oil (Cold Pressed)	1000	3.12	0.00
3	The Whole Truth Peanut Butter - Unsweetened & Crunchy	325	1.48	0.00
4	Popular Essentials Masoor Dal Split (Un-Polished)	500	0.63	0.00
5	Raw Banana	500	0.17	0.00
6	MTR Dal Makhani	300	1.10	0.00
7	Godrej Real Good Chicken Whole Pre-Cut	1000	2.70	0.00
8	Amul Spray Infant Milk Food	500	1.93	0.00
9	Baby Corn Peeled	250	0.29	0.00
10	Kitchens of India Dal Bukhara Gravy Mix	285	0.77	0.00
Total rows: 1395 Query complete 00:00:00.229				

Insight:

Products with lower price per gram provide better value to customers.

Product Weight Classification

Statement:

Categorized products into Low, Medium, and Bulk segments based on weight.

Output:

	name character varying (255)	weight_gms integer	weight_category text
1	Popular Essentials Poppy Seeds (Gasagase)	100	Low
2	Origami Kitchen Towels 2 In 1 Quilted 2ply	116	Low
3	Origami Good Karma Paper Serviettes - 100 napkins (1 ply)	58	Low
4	Johnson's Baby Top To Toe Bath Body Wash	210	Low
5	Harpic Hygienic Toilet Cleaner Rim Block Citrus	26	Low
6	Lizol Disinfectant Surface & Floor Cleaner Liquid Jasmine	500	Low
7	Sapota	348	Low
8	Britannia Treart Creme Wafers Chocolate	60	Low
9	Popular Essentials Dhania (Coriander Seeds)	200	Low
10	Popular Essentials Californian Almond	250	Low
Total rows: 1782 Query complete 00:00:00.181			

Total Inventory Weight by Category

Statement:

Calculated total inventory weight for each product category.

Output:

	category character varying (120)	total_weight bigint
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187
8	Personal Care	348187
9	Home & Cleaning	373161
10	Ice Cream & Desserts	490797
11	Chocolates & Candies	490797
12	Packaged Food	490797
13	Munchies	1404654
14	Cooking Essentials	1404654

Insight:

Useful for warehouse planning and inventory management.

7. SQL Concepts Used

- CREATE TABLE
 - SELECT
 - WHERE
 - ORDER BY
 - GROUP BY
 - HAVING
 - DISTINCT
 - CASE
 - Aggregate Functions
 - UPDATE
 - DELETE
 - ROUND
 - LIMIT
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8. Key Skills Demonstrated

- Data Cleaning
 - Exploratory Data Analysis (EDA)
 - Business Analytics
 - SQL Query Optimization
 - Aggregation & Reporting
 - Inventory Analysis
 - Pricing & Discount Analysis
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9. Key Findings

- Several products offer exceptionally high discounts, making them attractive for promotional campaigns.
 - Premium products that are out of stock could negatively impact potential revenue.
 - Certain categories contribute significantly more to overall revenue than others.
 - Price-per-gram analysis highlights products that deliver better value to customers.
 - Inventory weight analysis provides useful information for storage and logistics planning.
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10. Business Recommendations

- Maintain adequate stock levels for premium products with high demand.
 - Use high-discount products strategically to attract customers.
 - Focus marketing efforts on top revenue-generating categories.
 - Optimize pricing strategies for premium products with minimal discounts.
 - Utilize inventory analytics for efficient warehouse management.
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11. Conclusion

This project demonstrates the application of SQL in solving real-world retail business problems. Through data cleaning, exploratory analysis, and business-oriented querying, meaningful insights were extracted to support better decision-making. The project showcases practical SQL skills relevant to Data Analyst roles and forms an important part of a professional data analytics portfolio.